

# From Feedback to Insight: Understanding amazon Reviews through Text

The Amazon logo, a curved orange arrow pointing from the 'a' to the 'z', is positioned over the word 'amazon' in the title.

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## Our Dataset:

### Real-world Amazon customer data

- 6.4 million reviews
- Star ratings + review text

(<https://www.kaggle.com/datasets/sahityasahu/amazon-review-dataset/data>)

| overall | verified | reviewTime  | reviewerName     | reviewText                                                    | summary                   |
|---------|----------|-------------|------------------|---------------------------------------------------------------|---------------------------|
| 1       | FALSE    | 12 11, 2015 | Jacob M. Wessler | Alexa is not able to control my lights. If I ask her to te... | VERY Buggy, doesn't work. |
| 4       | FALSE    | 12 8, 2015  | Greg             | Alexa works great for me so far, but I'm also only con...     | So Far So Good            |
| 1       | FALSE    | 12 7, 2015  | Da-Gr8-1         | Weak!! Alexa doesn't even recognize the name Lifx. It'...     | Time waster               |
| 2       | FALSE    | 12 5, 2015  | Larry Russlin    | Can only control one of two bulbs from one of two ec...       | Buggy                     |
| 1       | FALSE    | 02 2, 2018  | Rebekah          | this worked great then randomly stopped. please upd...        | stopped working           |
| 5       | FALSE    | 01 15, 2018 | Nello            | Great skill                                                   | Great                     |
| 1       | FALSE    | 01 5, 2018  | Pete Johnson     | Pretty crappy. Won't connect with Alexis                      | Returning to              |
| 1       | FALSE    | 01 4, 2018  | L. Ray Humphreys | Not happy. Can not connect to Alexa regardless.               | Can not connect to ECHO   |
| 1       | FALSE    | 12 30, 2017 | Viola            | Can not connect a hue lights to Alexa. Linked the LIFX...     | Connecting is a no go     |
| 1       | FALSE    | 12 29, 2017 | angie anj        | The service works with google home, but doesn't wor...        | Does not work             |
| 5       | FALSE    | 12 25, 2017 | Amazon Customer  | So easy to connect and I'm having way too mu...               | This is awesome           |
| 5       | FALSE    | 12 18, 2017 | Artyhow          | I use "Echo" as the wake word. I say &I...                    | Alexa and LIFX            |

Summary: Call a Taxi

Overall: ★☆☆☆☆

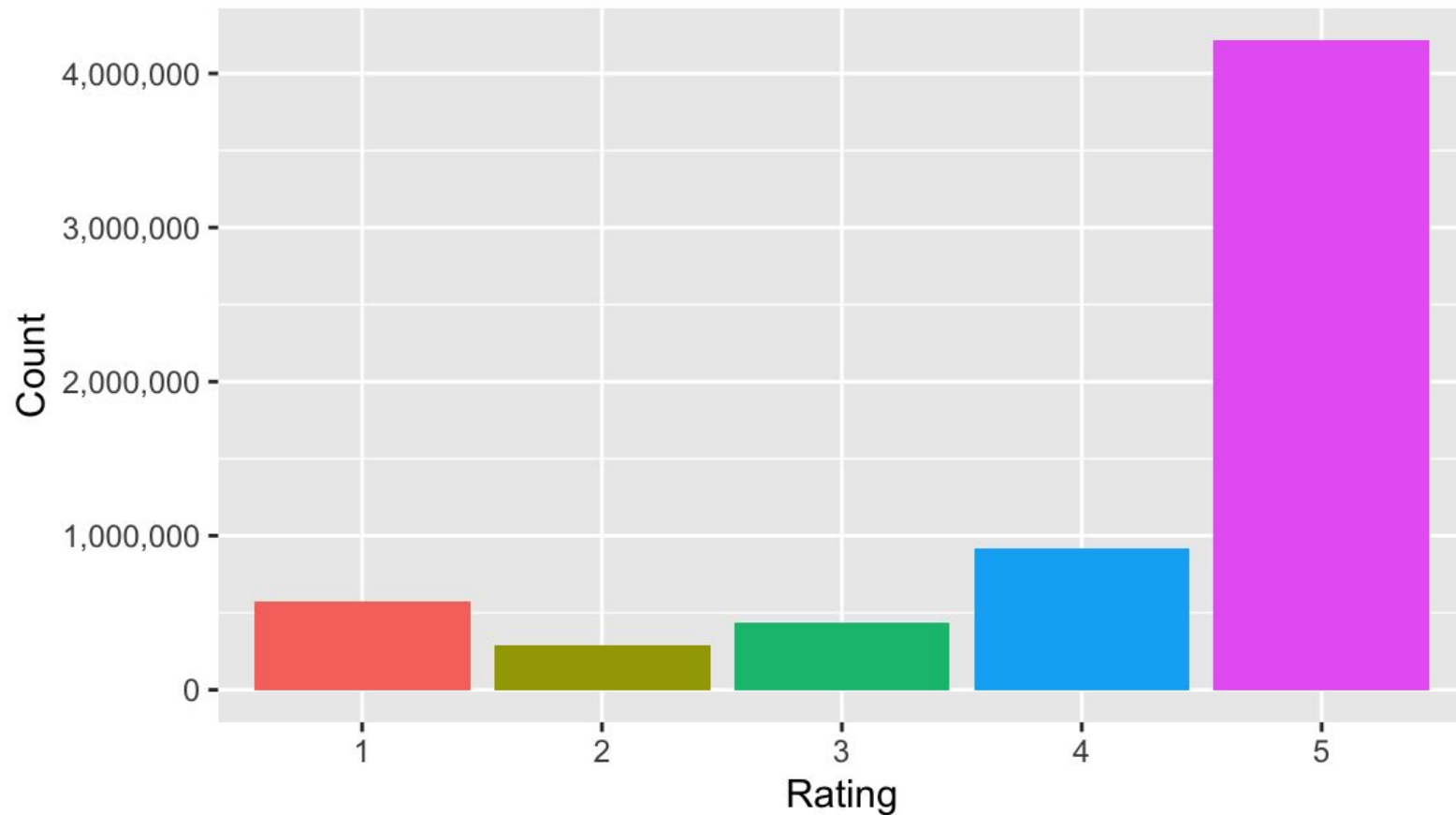
Review: Zero stars. Don't work. I request a ride all kind of ways and it put my request to my shopping list. What a joke. Called a cab  
-Glenn Smith

Summary:Garbage

Overall: ★☆☆☆☆

Review: You'd think Amazon would do a little research on how horrible AccuWeather is but clearsell outs happen every day.  
-Amazon Customer

# Distribution of Review Ratings in the Full Amazon Dataset



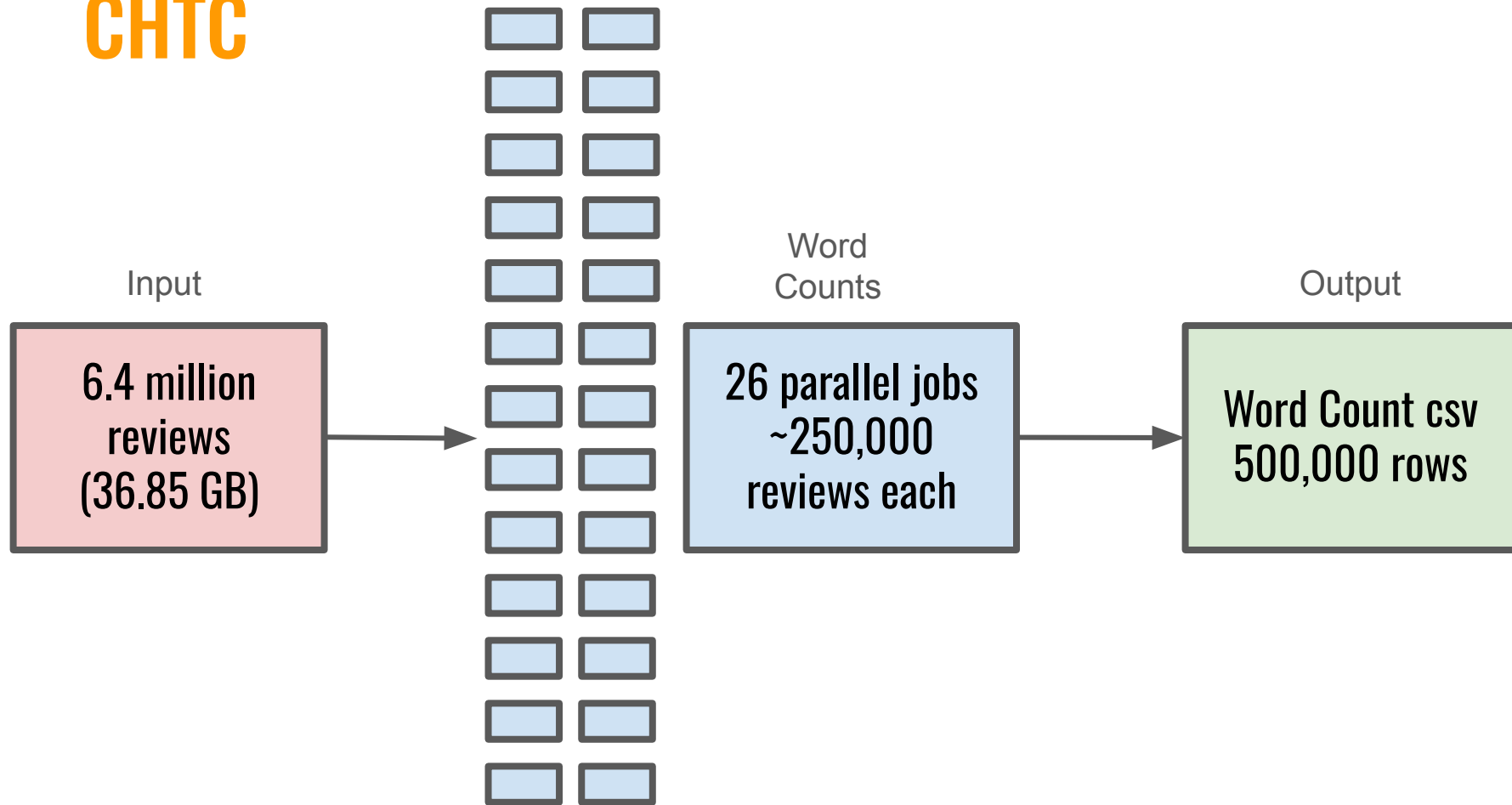
# Target Audience:

Business and product teams looking to improve their products based on customer reviews and identify what customers love most to guide marketing strategies

## **Research Question:**

**What words and language patterns are most associated with low-rated versus high-rated Amazon reviews?**

# CHTC

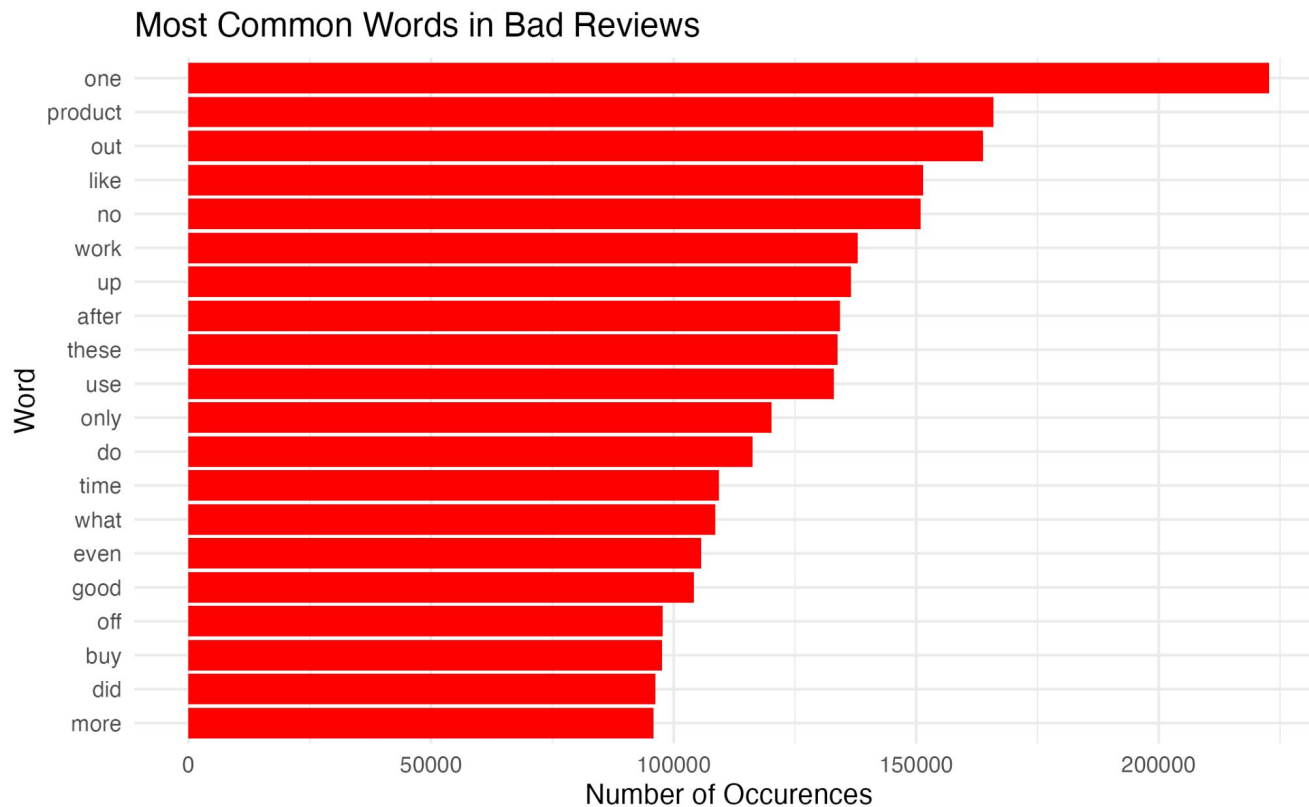


# Word Counts





# Word Counts



# Data Cleaning

- Extracted reviewText field from the Amazon review JSON data
- Filtered reviews into two groups:
  - Low rated: 1-2 stars
  - High rated: 4-5 stars
- Removed 3-star reviews (neutral)
- Removed punctuation and tokenized text into individual words
- Removed blank entries
- Removed common words such as “the”, “and”, “is”, and “to”
- Computed word frequencies for low and high rated reviews separately

# Our Models

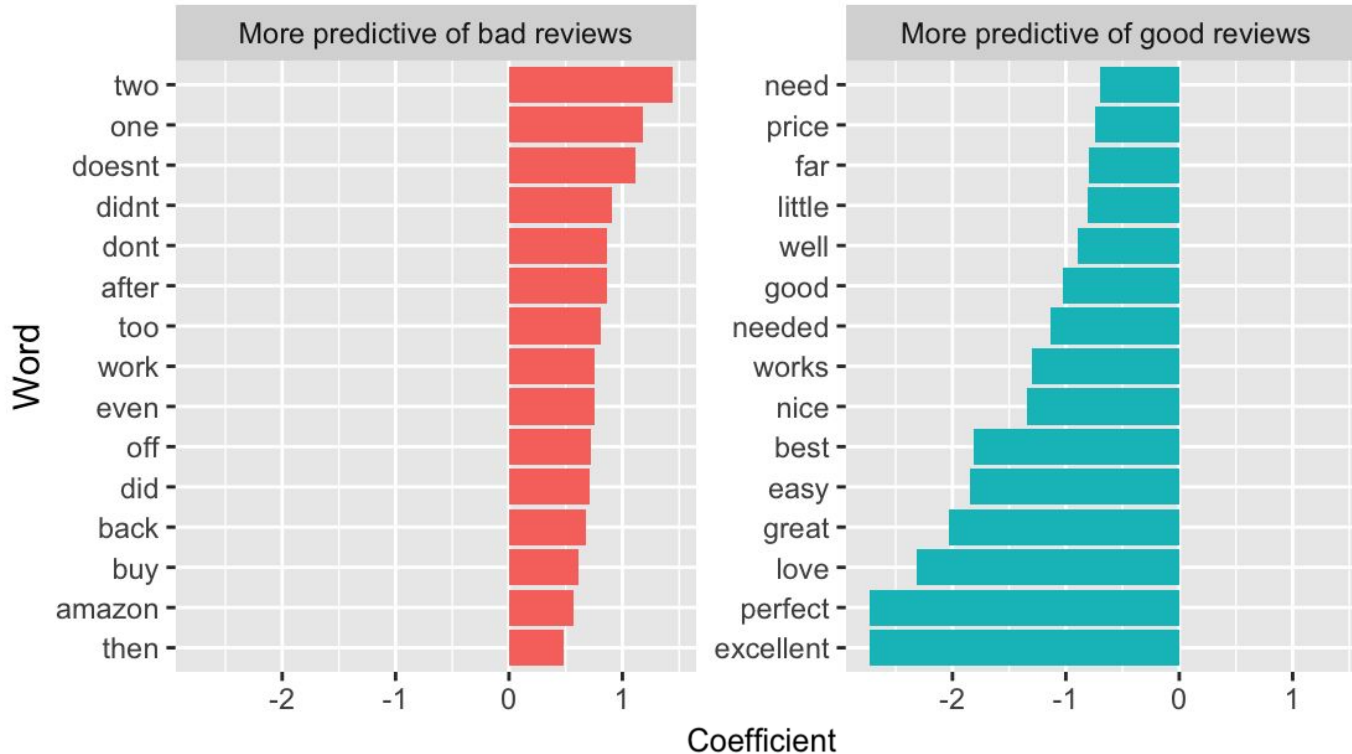
1. Logistic Regression
2. Random Forest

$X$  = Binary Word indicators  
from the top 100 most  
common words

$y$  = High-Rated, Low-Rated

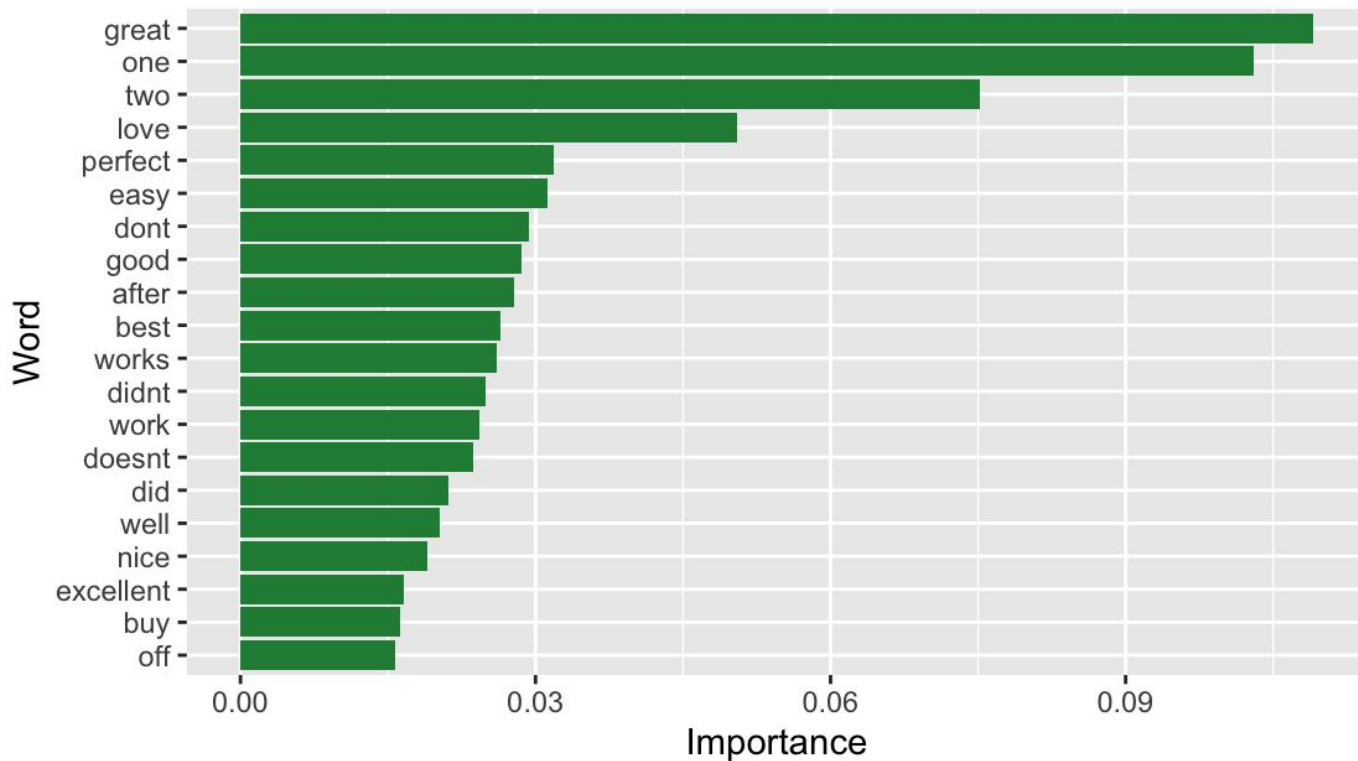
# Logistic Regression

## Most Influential Logistic Regression Words



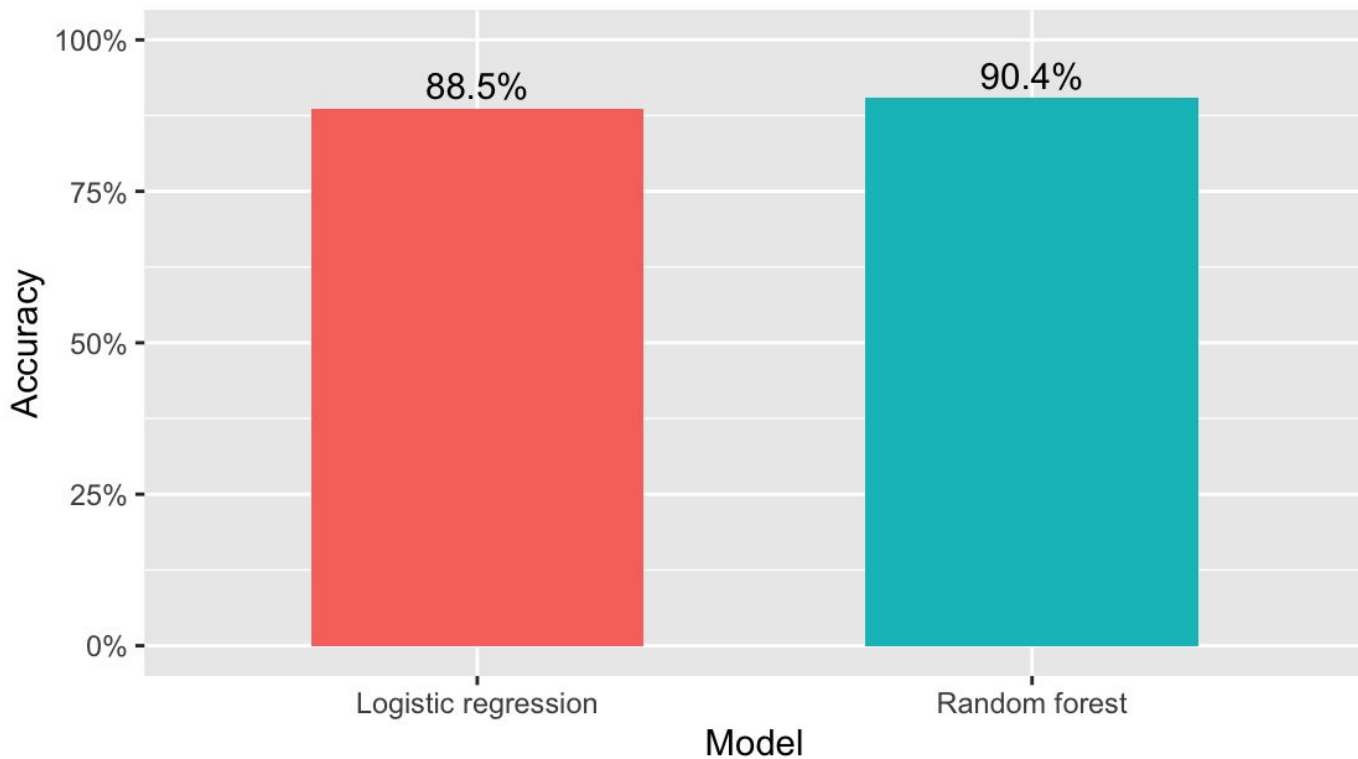
# Random Forest

Top Random Forest Word Importance Scores



# Accuracy

Model Accuracy on Full Amazon Review Test Set



# Results and Conclusions

High-rated reviews had clear and consistent language patterns, with terms like “excellent”, “perfect”, and “love” strongly tied to positive customer experiences

Negative Reviews commonly mention malfunctions (doesn't, work, off)

# Challenges

- The dataset was imbalanced with more high-rated reviews than low-rated reviews
- The model uses individual words, so they lost phrases like “doesn’t work” or “don’t buy”
- Some words were hard to interpret because the model could not tell how they were being used in a sentence.



# Future Considerations

- Using two word phrases to capture clearer patterns
  - Better measure of customer opinion
- Use a more balanced dataset

**Thank you for listening!**